



ISI MSQE PEA 2022 PYQ Practice Booklet

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STATSRIVE

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STATSTRIVE

Question 1

Consider an economy with two goods X and Y . Let the utility function be given by $u(x, y) = A\sqrt{xy}$ where $A > 0$, $x \geq 0$ and $y \geq 0$. The budget constraint is $P_X x + P_Y y \leq M$, with $M > 0$. Let $P_X = P_Y > 1$ and let (x^*, y^*) denote the utility maximizing bundle. Then,

- (A) it must always be that $x^* > y^*$
 - (B) it must always be that $x^* = y^*$
 - (C) it must always be that $x^* < y^*$
 - (D) it must always be that $x^* + y^* = M$
-

Question 2

For the linear utility $u(x_1, x_2) = 3x_1 + 2x_2$ with budget $2x_1 + 3x_2 \leq M$, $x_1, x_2 \geq 0$, the Lagrangian is

$$L(x_1, x_2; \lambda) = 3x_1 + 2x_2 + \lambda[M - 2x_1 - 3x_2].$$

Determine the equilibrium $(x_1^*, x_2^*, \lambda^*)$.

- (A) $(x_1^* = M/2, x_2^* = 0, \lambda^* = 3/2)$
 - (B) $(x_1^* = 0, x_2^* = M/3, \lambda^* = 2/3)$
 - (C) $(x_1^* = M/2, x_2^* = 0, \lambda^* = 2/3)$
 - (D) $(x_1^* = 0, x_2^* = M/3, \lambda^* = 3/2)$
-

Question 3

In month 1: $p_X = 2, p_Y = 3$. Consumer A chose $(3, 8)$, consumer B chose $(6, 6)$. In month 2: $p_X = 3, p_Y = 2$. Consumer A chose $(8, 3)$, consumer B chose $(4, 9)$. Which statement is correct?

- (A) Both consumers satisfy WARP
 - (B) Neither consumer satisfies WARP
 - (C) Consumer A satisfies WARP but not consumer B
 - (D) Consumer B satisfies WARP but not consumer A
-

Question 4

Production function: $Y(L, K) = \min\{2L, K\}$; cost $C = wL + rK$, $w, r > 0$. Find (L^*, K^*) minimizing cost subject to $Y(L, K) \geq \bar{Y}$.

- (A) $L^* = \bar{Y}$ and $K^* = \bar{Y}/2$
 - (B) $L^* = \bar{Y}$ and $K^* = \bar{Y}$
 - (C) $L^* = \bar{Y}/2$ and $K^* = \bar{Y}$
 - (D) None of the other options is correct
-

Question 5

Inverse demand $P = 1 - q_1 - q_2$; cost $c_i(q_i) = \kappa_i q_i$, $\kappa_i \in (0, 1)$. Find firm 2's Cournot equilibrium profit.

- (A) $(1 - \kappa_1 + \kappa_2)^2/9$
 - (B) $(1 - \kappa_2 + \kappa_1)^2/9$ [i.e. $(1 - 2\kappa_2 + \kappa_1)^2/9$ corrected below]
 - (C) $(1 - 2\kappa_1 + \kappa_2)^2/9$
 - (D) $(1 - 2\kappa_2 + \kappa_1)^2/9$
-

Question 6

Which statement is correct in a two-good world?

- (A) Diminishing MU of both goods is sufficient for diminishing MRS
 - (B) Diminishing MU of both goods is necessary for diminishing MRS
 - (C) Diminishing MU of at least one good is necessary for diminishing MRS
 - (D) Diminishing MU of at least one good is neither necessary nor sufficient for diminishing MRS
-

Question 7

A non-transitive preference relation can be represented by a utility function

- (A) Always
 - (B) Only if preferences are complete
 - (C) Only if preferences are complete and convex
 - (D) Never
-

Question 8

$U(x, y) = \min\{x, y\}$, income $M = 200$. Old prices $(p_x, p_y) = (2, 2)$; new prices $(2, 3)$. Find the equivalent variation A and compensating variation B .

- (A) $A = 30, B = 70$
 - (B) $A = 40, B = 50$
 - (C) $A = 50, B = 75$
 - (D) $A = 60, B = 60$
-

Question 9

$U(x, y) = -[(10 - x)^2 + (10 - y)^2]$. All prices equal 1, income = 40. Optimal (x, y) ?

- (A) $(10, 10)$
 - (B) $(0, 0)$
 - (C) $(5, 5)$
 - (D) None of these
-

Question 10

$F(K, L) = \min\{aK, bL\}$, $a, b > 0$, $a \neq b$. For fixed $\bar{K}$, the marginal product of labor is

- (A) 0
 - (B) $1/a$ if $L < (a/b)\bar{K}$ and 0 otherwise
 - (C) $1/b$ if $L < (b/a)\bar{K}$ and 0 if $L > (b/a)\bar{K}$
 - (D) None of the above
-

Question 11

Let $e_i(p_0)$ be the price elasticity of demand for good X of consumer i at p_0 , with heterogeneous quantities. What is the elasticity of the aggregate demand at p_0 ?

- (A) $\sum e_i(p_0)$
 - (B) $\sum \frac{q_i(p_0)}{\sum_j q_j(p_0)} e_i(p_0)$
 - (C) $\sum \frac{1}{N} e_i(p_0)$
 - (D) None of these
-

Question 12

m identical firms with $C(q) = q^2 + 1$. Industry demand $D(P) = a - bP$, $a, b > 0$. Short-run equilibrium output per firm?

- (A) 0
 - (B) $a/(m + 2b)$
 - (C) $a/(m^2 + b)$
 - (D) $a/(m + b/2)$
-

Question 13

$C(q) = 3q^2 + 800$; $P = 280 - 4q$. Find the price elasticity of demand at the profit-maximizing price.

- (A) -4.5
 - (B) -3.5
 - (C) -2.5
 - (D) -1.5
-

Question 14

Solow model with saving s , depreciation δ , labor growth n , no technological progress. Steady-state capital-output ratio?

- (A) $s/(n + \delta)$
 - (B) $n/(\delta + n)$
 - (C) $\delta/(s + n)$
 - (D) $1/(s + n + \delta)$
-

Question 15

Number of arrangements of the word “PANDEMIC” (8 distinct letters) such that the vowels appear together.

- (A) $6 \times (3!)(5!)$
 - (B) $5 \times (3!)(5!)$
 - (C) $4 \times (3!)(5!)$
 - (D) $1 \times (3!)(5!)$
-

Question 16

$f(x) = x^2 - x - 1$, $g(x) = x + 1$. Let $\alpha_1 > 0$ and $\alpha_2 < 0$ be the roots of $f(x) = 0$; let $\beta_1 > 0$ and $\beta_2 < 0$ be the roots of $f(g(x)) = 0$. Identify which statement is *incorrect*.

- (A) $\alpha_1 - \beta_1 = \alpha_2 - \beta_2 = 1$
 - (B) $\alpha_1 + \beta_2 = \alpha_2 + \beta_1 = 0.5$ [as printed]
 - (C) $\alpha_1 + \beta_1 = -(\alpha_2 + \beta_2) = \sqrt{5}$
 - (D) $\alpha_1 + \alpha_2 = -(\beta_1 + \beta_2) = -1$
-

Question 17

$f(x) = \sqrt{1-x}$. Then $f(1-f(x))$ equals

- (A) x
 - (B) $\sqrt{1-x}$
 - (C) x^2
 - (D) $1-x^2$
-

Question 18

f is increasing, concave, C^2 ; g is decreasing, convex, C^2 . Then $G(x) = g(f(x))$ is

- (A) increasing and convex
 - (B) decreasing and convex
 - (C) increasing and concave
 - (D) decreasing and concave
-

Question 19

$f : \mathbb{R} \rightarrow \mathbb{R}$ is C^2 with $f''(x) > 0$ for all x , $f(1) = 1$, $f(2) = 2$. Then,

- (A) $0 < f'(2) < 1$
 - (B) $f'(2) > 1$
 - (C) $f'(2) = 1$
 - (D) $f'(2) = 0$
-

Question 20

A, B non-singular of the same order, $C = BAB^{-1}$. For scalar λ , $\det(C + \lambda I)$ equals

- (A) $\det A$
 - (B) $\det B$
 - (C) $\det(A + \lambda I)$
 - (D) $\det(B + \lambda I)$
-

Question 21

$$\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}.$$

- (A) 0
 - (B) e
 - (C) $1/e$
 - (D) None of these
-

Question 22

$f : [0, \infty) \rightarrow \mathbb{R}$, $f(0) = 0$, $f''(x) > 0$ for $x > 0$. Then $f(x)/x$ is

- (A) increasing in $(0, \infty)$
 - (B) decreasing in $(0, \infty)$
 - (C) increasing in $(0, 1]$ and decreasing in $(1, \infty)$
 - (D) decreasing in $(0, 1]$ and increasing in $(1, \infty)$
-

Question 23

$f : A \rightarrow B$ with $|A| = 5$, $|B| = 2$. How many onto functions?

- (A) $5^2 - 1$
 - (B) $5^2 - 2$
 - (C) $2^5 - 1$
 - (D) $2^5 - 2$
-

Question 24

$f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{x}{1+x^2}$. Then

- (A) $-1 \leq f(x) \leq 1$
 - (B) $-1 \leq f(x) \leq 1/2$
 - (C) $-1/2 \leq f(x) \leq 1$
 - (D) $-1/2 \leq f(x) \leq 1/2$
-

Question 25

A is 3×3 with rank 3; B is 3×4 with rank 3. Then $\text{rank}(AB)$ is

- (A) 3
 - (B) 4
 - (C) 6
 - (D) 7
-

Question 26

Let A be a specified 6×8 (or square, source unclear) matrix [the entries of A were not legibly transcribed in the source]. Which one is an eigenvalue of A ?

- (A) 2
 - (B) 1
 - (C) 3
 - (D) 5
-

Question 27

A is a 5×5 non-null singular matrix. Then $Ax = 0$ has

- (A) only the trivial solution
 - (B) exactly 5 solutions
 - (C) no solution
 - (D) infinitely many solutions
-

Question 28

A family has two children. Find the probability that both are boys given that at least one is a boy.

- (A) $1/2$
 - (B) $2/3$
 - (C) $1/3$
 - (D) $1/4$
-

Question 29

Box 1 contains {1 black, 1 white}; Box 2 contains {1 black, 2 white}. Select a box at random and then a ball at random. Probability that the drawn ball is black?

- (A) $5/12$
 - (B) $2/5$
 - (C) $1/6$
 - (D) $5/11$
-

Question 30

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = (x^2 + 1)^{2022}$.

- (A) one-one but not onto
 - (B) onto but not one-one
 - (C) both one-one and onto
 - (D) neither one-one nor onto
-